

Mem0 paired evaluation update

Run date: 27 July 2026 · DeepSeek V4 Flash · seed 42 · top-5 retrieval

This report replaces the earlier partial-session LoCoMo pilot. Both backends ingested all 19 dated sessions of conv-26 before answering the same 40 questions.

Important comparability note: these are results for this repository's DeepSeek, session-level benchmark configuration—not a reproduction of the Mem0 paper. The paper uses GPT-4o-mini and incremental message-pair extraction with a rolling 10-message context, an asynchronously refreshed conversation summary, top-10 similar memories during updates, and a two-speaker timestamp-aware answer prompt. Do not compare the F1 values here directly with the paper's published Mem0 scores.

What was compared

Text memory

2.0.14

Current Mem0 with local Qdrant vector retrieval.

Graph memory

0.1.45

Historical Mem0 MemoryGraph with Neo4j.

Paired samples

60 / 40

Identical PersonaMem rows and LoCoMo questions for both backends.

Version limitation: this compares complete current and historical implementations, not a pure vector-versus-graph architecture ablation.

Memory changes: 15 controlled updates

Every case adds an original fact, adds a conflicting replacement, searches, inspects storage, then deletes its isolated scope. Exact identifiers make stale facts auditable.

Metric	Text	Graph
New fact retrieved after update	15/15	12/15
Old fact retrieved after update	15/15	4/15
Old fact still stored after update	15/15	11/15
Clean after explicit deletion	15/15	15/15
Retrieval p50	22 ms	1,674 ms
Operation p50	1.49 s	7.52 s

Graph suppressed stale facts more often in top-5 retrieval, but frequently retained stale graph state and sometimes missed the current fact. Text always retrieved the current fact but retained every obsolete fact.

PersonaMem-v2: 60 independent conversations

Each row stored only its own related conversation snippet, then used the same answer model with and without retrieved memory.

Metric	Text	Graph
MCQ accuracy	29/60 · 48.3%	23/60 · 38.3%
No-memory accuracy	13/60 · 21.7%	13/60 · 21.7%
Memory improvement	+26.7 points	+16.7 points
Non-empty retrieval	76.7%	66.7%
Updated/forget cases correct	2/10	0/10

Metric	Text	Graph
Ingestion / retrieval p50	2.14 s / 29 ms	10.90 s / 1,715 ms

The direct text-minus-graph accuracy difference was +10.0 points, with a paired 95% interval of −1.7 to +21.7 points. Both systems improved over their no-memory controls, but this sample does not establish a conclusive direct accuracy win for text.

LoCoMo: one complete 19-session conversation

conv-26 was fully ingested by both backends. The same 40 questions were evaluated: 10 each from single-hop, temporal, multi-hop, and open-domain categories. Category 5 was excluded.

Each dated session is submitted as one ingestion unit in this benchmark. That preserves complete-session context, but differs from the paper’s incremental message-pair memory pipeline.

Metric	Text	Graph
Official LoCoMo F1	13.96%	3.31%
No-memory F1	0.81%	0.81%
Memory F1 improvement	+13.15 points	+2.50 points
Answer-token recall	50.62%	4.88%
Answer abstention	42.5%	97.5%
Retrieval / session-ingestion p50	35 ms / 4.60 s	1,421 ms / 17.56 s

Question type	Text F1	Graph F1
Single-hop	7.86%	10.00%
Temporal	27.46%	0.00%
Multi-hop	4.66%	2.00%

Question type	Text F1	Graph F1
Open-domain	15.87%	1.25%

The paired text-minus-graph F1 difference was **+10.65 points**, 95% interval **+3.01 to +17.52**. Both systems returned non-empty retrieval for every question, but graph content rarely supported an answer.

Interpretation

- Text memory was faster and stronger on this complete-conversation LoCoMo run.
- Graph reduced stale-fact retrieval in controlled updates, but stale graph state and current-fact misses remained important failures.
- Historical graph processing logged malformed deletion-extraction responses and search-tool errors during PersonaMem, although all 60 rows produced reports.
- These are paired pilots, not a reproduction of published Mem0 results: the model, ingestion unit, extraction context, update retrieval depth, and answer prompt differ from the paper. More complete conversations and repeated seeds are needed before generalizing.

Raw reports

[Memory changes — text](#) · [Memory changes — graph](#) · [PersonaMem — text](#) · [PersonaMem — graph](#) · [LoCoMo — text](#) · [LoCoMo — graph](#)